Geochemical study of Amoeboid Olivine Aggregates in Mukundpura CM and Allende CV Chondrites: Insights from EPMA Analyses

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CERTIFICATE

Mr. Rishant Prakash, MTech 1st yr student at Centre for Earth Sciences of Indian Institute of Science, Bengaluru has successfully completed his Summer Internship Project-2022 at Physical Research Laboratory, Ahmedabad with the title ***"Geochemical study of Amoeboid Olivine Aggregates in Mukundpura CM and Allende CV Chondrites: Insights from EPMA Analyses"*** under my supervision.

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Chapter-1

INTRODUCTION

The most widely accepted nebular hypothesis for the formation of our solar system suggests that the gravitational collapse of a giant molecular cloud fragment led to the formation of the proto-sun at its centre and a rotating disk of gas and dust, called the solar-nebula, surrounding the nascent sun (Bhattacharya & Litchman, 2017). The solar system objects (planets, satellites, comets, and asteroids) formed out of this nebula. Planets are by-products of star formation and are formed from the solar nebula, the disk-shaped cloud of gas and dust left over from the Sun's formation (Goldreich & Ward, 1973).

A meteorite is defined as an object that has an extra-terrestrial or outer space origin and have been found or collected on the surface of the Earth. Based on the origin and evolution of meteorite, they are classified as undifferentiated and differentiated. All the chondritic meteorite belongs to undifferentiated meteorite while achondrite meteorites are in differentiated group. Chondrites as per its name, consists of molten ferromagnesian spherical shape chondrule, matrix, refractory inclusion, metal (Fe-Ni) etc. Depending on the chemistry, mineralogy, petrography, and oxygen isotopes, chondrites are divided into 4 groups (1) Carbonaceous chondrite (CC) (2) Ordinary chondrite (OC) (3) Rumuruti chondrites (4) Enstatite Chondrite. Carbonaceous chondrites are primitive and mostly located in the outer part of the asteroid belt. There are eight well-resolved groups of carbonaceous chondrites: CI, CM, CR, CH, CB, CV, CK, and CO. The detailed classification of meteorites is shown in Fig (1).

The study of Carbonaceous chondrite also provides insight into different parent body processes like aqueous alteration, thermal alteration, and shock metamorphism. CI, CM and CR meteorites have passed through aqueous alteration. Based on the formation, the total volume percentage of secondary minerals, mainly the phyllosilicates and poorly characterised phase (PCP), the alteration in CM meteorites can further be divided into different grades, i.e., CM2.6 to CM2.1, where 2.6 is the least altered phase. Like other chondritic meteorites, CM type also consists of chondrule (either unaltered, partially altered or completely altered), fine-grained matrix, metal (Fe-Ni and Fe-Ni-S), and refractory inclusion (Ca-Al rich inclusion (CAIs) and amoeboid olivine aggregates (AOA)). Four main parts make up chondrites: chondrules, Fe-Ni metal, refractory inclusions (Ca-Al rich inclusion (CAIs) and amoeboid olivine aggregates (AOAs)) and a fine-grained matrix that is richer in volatile elements (Krot. et al. 2022)

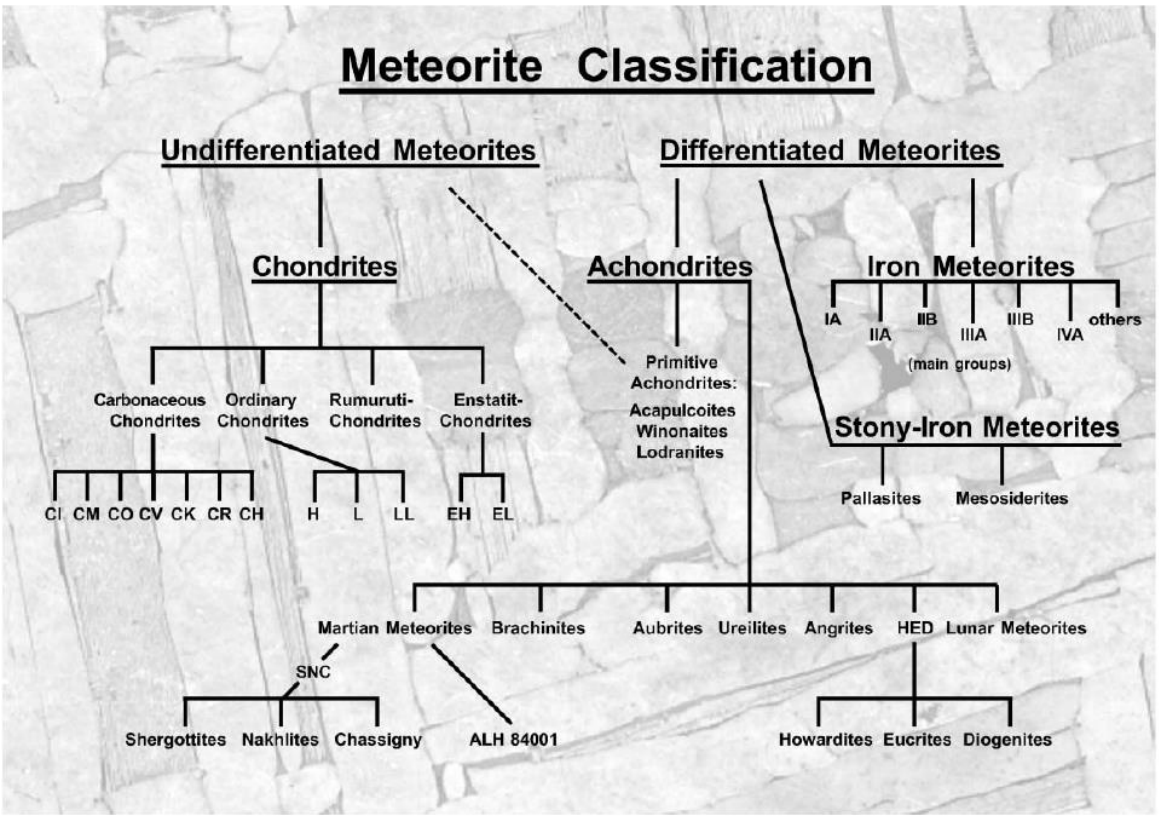


Figure 1: Diagram expressing the systematics of meteorite classification and showing the major meteorite division, classes, and groups

Amoeboid olivine aggregates (AOAs) are irregular shaped refractory inclusion which are predominantly composed of olivine grains. Grossman and Steele (1976) based on the texture have hypothesised that AOAs are primary solid condensates from the solar nebula that holds the information of about the early solar nebula. Olivine being the most predominant mineral in the chondritic meteorites, AOAs then become one of the best samples to study the development of solid material in the early solar system.

AOAs are irregularly shaped aggregates of mainly three components, forsterite olivine (Fo_{96} for Mukundpura and $Fo_{>96}$ for Allende), fine grained altered CAIs and Fe-Ni inclusions. In general AOAs in all carbonaceous chondrites show almost similar texture, mineralogy and isotopic composition. The mean size of AOAs vary among the chondrite groups for example CM2- 225 μm, LL3 200 μm, CV3 3mm (Rubin 2013) and CK 900 μm. The AOAs are considered genetically related to CAIs as the minerals present in AOAs are derived from early high temperature condensates of Ca-Al rich environment and chemically very similar to magnesium type-I chondrule. Hence, the study of AOAs is important as it forms a genetic link between CAIs and chondrules. This study mainly focused on the mineralogical composition of AOA in CM meteorite and CV meteorite.

Chapter-2

SAMPLE USED IN THIS STUDY

The two samples used for the geochemical study of Amoeboid olivine aggregates are CV chondrite (Allende) and CM chondrite (Mukundpura). The literature study describing the mineralogical characteristics of these samples is discussed below.

2.1Mineralogical Characteristics of CV and CM chondrites

CV Chondrite

The CV chondrites are a diverse group of chondritic meteorites which was split into reduced (CVR) and oxidised (CVOx) subgroups based on the modal metal/magnetite ratios and nickel content of metal and sulphides (McSween, 1977). The oxidised CV chondrites are subdivided into Allende -like (CVOxA) and Bali-like (CVOxB) subgroups (Weisberg et al. (1997). It was observed that the Metal/magnetite ratios decrease in the same order as the matrix/chondrule ratios, which increase in the sequence CVR(0.5-0.6), CVOxA(0.6-0.7), and CV3OxB(0.7-1.2) (McSween,1997; Simon et al., 1995). The CV chondrites are characterised by (a) high matrix/chondrule ratios (0.5-1.2); (b) chondrules of sizes in millimetres with mostly showing porphyritic textures, most of which are Mg-rich and 50% of that have course-grained igneous rims around it; (c)high abundances of CAIs and AOAs of mm-cm size; and (d) common occurrence of igneous melilite-spinel-pyroxene ± anorthite. Oxygen isotope compositions of the CV chondrite show a wide range. The CVOxB chondrites are depleted in 16O compared to CVOxA and CVR chondrites (Weisberg et al., 1997). Over time, a significant mineralogical difference has been observed in CV subgroups. These differences studied are mainly secondary and have resulted from late-stage alteration (Krot et al., 1995, 1998). For instance, phyllosilicates (abundant in CVOxB), magnetite, Ni-rich sulphides, hedenbergite and fayaltic olivine (Fa95-100) replace the primary minerals in chondrules (Hua and Buseck, 1995; Krot et al., 1998) whereas magnetite, Ni-rich sulphides, fayalitic olivine (Fa30-60) replaces the primary minerals in CVOxA (Allende) and chondrules in CVR chondrites remain unaltered (Kimura and Ikeda, 1998; Krot et al., 1995,1998). Phyllosilicates are uncommon in both the CVR and CVOxA chondrites, despite the fact that all CV chondrites have been categorised as type 3 (CV3); they are prevalent in the matrix of CVOxB chondrites. (Kellar and McKay, 1993; Kellar et al., 1994; Kellar and Buseck, 1990, Tomeoka and Buseck, 1982; Zolensky et al., 1993).

The materials in CV -chondrite matrices contain fragmented pieces of chondrule, CAIs and AOAs debris (including whole minerals and fragments of minerals). Troilite, Low Ca-Pyroxene, olivine, anorthite, spinel, Ca-Mg rich Pyroxene (diopside) and Al-Ti diopside (fassaite) are the phases that are derived from chondrule, CAI and AOA fragments into the matrices. Phases formed by secondary processes involve aqueous/hydrothermal alteration. And the minerals most likely derived from secondary processes include fosteritic and fayalitic olivine, monticellite,

kirschsteinite, hedenbergite, wollastonite, nepheline, sodalite, andradite magnetite, ilmenite, wadalite, Al-rich serpentine, Na-K mica, and calcite. The primary phases in the matrices include troilite, olivine, and low-Ca Pyroxene. These minerals may have both primary and secondary origins in CV matrices.

The CVOxA subgroup chondrites comprise very little H2O(<0.2 wt% in the Allende fall) and show rare phyllosilicates in their matrices. But different phyllosilicate minerals such as phlogophite; montmorillonite, clintonite, margarite, saponite, and chlorite-group minerals are still present. Fe Olivine is the most abundant phase found in the Allende matrix.

CM chondrites

The sample other than Allende used for analysing AOA is Mukundpura chondrite. A very recent fall was observed on 6th June 2017 in Mukundpura, Rajasthan. The entire fragmented fall pieces collected weighed 2 kg. It was acquired soon after the fall to avoid terrestrial contamination (Tripathi et al., 2018; Rudraswami et al., 2018; Ray and Shukla, 2018).

CM (Mighei-like) chondrites are well established from other carbonaceous chondrite groups based on the oxygen isotopic compositions and bulk chemistry. The major petrographic and mineralogic descriptions include (a) chondrules (relatively small in size (<300mm)) of various textural types (majorly porphyritic), chondrules in CM chondrites comprises <30 vol% and are partially replaced by phyllosilicates; (b) many chondrules show fine-grained accretionary rims around them; (c) due to extensive alteration the presence of Fe-Ni metal is almost nil; (d) commonly found CAIs and AOAs; and (e) matrix composed of phyllosilicates, tochilinite, sulfides, carbonates, and magnetite.

The matrix of CM-type chondrites comprises Fe-Ni (kamacite) that decreases in their availability during aqueous alteration. Chondrites often contain 0.02 vol% metallic Fe-Ni. The altered CM chondrites consist of many sulfide grains with "intermediate compositions" (atomic Ni/(Ni+Fe) ratios of 0.1-0.40. The aqueous alteration in CM chondrites spans a large range, varying from type 2.0, which is highly altered, to 2.7 to 2.9. Rubin et al. (2007 and 2015) used a system for classifying CM chondrites based on the degree of aqueous alteration. Fe-Mg serpentines of varying composition are abundantly found in CM Chondrites. The MgO/FeO ratios show increasing sequence with the increase in their degree of alteration. The least altered (type 2.0) show amorphous silicate materials, whereas the highly altered (type 2.7 or 2.9) show abundant phyllosilicates (Rubin et al. 2006).
The mineralogy and petrography study of Mukundpura chondrite showed the presence of abundant phyllosilicates, fine-grained accretionary rim surrounding chondrules (Potin et al., 2020; Ray and Shukla, 2018; Rudraswami et al. 2018). The petrographic grade 2 was assigned to the chondrite, given it didn't lose much of its pristine nature (petrographic grade 1 if lost) and showed no signs of thermal alteration (petrographic grade 3) because of the presence of abundant phyllosilicates in

the matrix. Also, the FeO/SiO2 (2.7) and S/SiO2 (0.4) matched with Murchison, which is another CM2 chondrite (type 2.5) (FeO/SiO2 =2.7 and S/SiO2= 0.35) (Ray and Shukla, 2018).

In this study, we present the results of the geochemical analysis of AOAs in CM 2.5 (Mukundpura) and CV3 (Allende). Based on the mineral chemistry of the AOAs and related objects, we have tried to understand the formation sequence of minerals, interpret their origin and understand their relationship with other chondritic components in CM and CV3 chondrites.

Chapter -3

METHODOLOGY

The analytical methods applied in this study include Energy Dispersive X-Ray Spectroscopy (EDS) and Electron microprobe analyser. The qualitative analysis of chemical composition can be obtained using an Energy Dispersive X-Ray Spectroscopy (EDS) technique. This instrument in general comes as an attachment with SEM or Electron micro probe analyser. By principle both EDS and EPMA measure the composition from X-ray intensity of elements. EPMA used WDS (Wavelength Dispersive Spectroscopy) technique.



Figure 2: JEOL 8530 FE-EPMA (Field Emission Electron Micro Probe Analyzer) based at PRL, Ahmedabad

The meteoritic samples taken were one Allende (CV3) and two Mukundpura (CM2) of size <1cm. They were first mounted using the epoxy solution. And it was left overnight to dry. The prepared thick section was then taken for polishing. The powder made of corundum of varying diameter from 1 μm to 0.3 μm is used for the polishing of the sample. The mount gets polished subsequently with corundum powder of different sizes in diameter until the sample gets exposed, i.e., the sample starts coming on the surface. The process was followed in a stepwise manner depending on the level of exposure required on the surface of the mount. The smoother the surface more precise the result we get during the analysis of our sample under EPMA. The geochemical analyses, including petrographic characterisation, textural observation, back-scattered electron (BSE) imaging, and data collection, were performed using a field emission electron prole microanalyser (FE-EPMA; JEOL 8530F, Japan) (fig 2) with Energy Dispersive X-ray Spectrometer (EDS) and five Wavelength Dispersion Spectrometer (WDS).

EPMA stands for Electron Probe Micro Analyzer, it is a microbeam instrument used specifically for in-situ non-destructive chemical analysis of a minute-sized sample. EPMA works by bombarding a micro-volume of a sample with a focused electron beam (typical energy = 5-30 keV) and collecting the X-ray photons thereby emitted by the various elemental species. Here for our study, the minerals were analysed, and the images were taken using a typical accelerating voltage of 15keV and 15 nA sample current and beam size of 1 micron. Because the wavelengths of these X-rays are characteristic of the emitting species, the sample composition was first easily identified by recording WDS spectra, and then EDS data were collected for further calculation. WDS spectrometers are based on Bragg's law and use various moveable, shaped monocrystals as monochromators. Routine quantification to 1% reproducibility is obtained over several days. It is the most precise and accurate micro-analysis technique available, and all elements from B to U and above can be analysed. For the silicate mineral, the element was calibrated with silicates standard (Na- Jadeite, Mg –Olivine, Si-Diopside, Al-kyanite, P-Apatite, K-Orthoclase, Ca-Wollastonite, Ti-TiO2 (Rutile), Cr- Cr_2O_3, Mn-Rhodonite, Fe-Andesite Ni- Ni metal, Co-Co metal. For metal analysis, elements were calibrated with metal standards (Fe-Fe metal, Ni-Ni metal, Co-Co metal, Cr-Cr metal and S-pyrite)

Chapter-4

RESULTS

Textures

The AOAs observed in the Allende (CV3) and Mukundpura (CM 2.5) are irregular in shape and size, approximately 900 µm and 150-200 µm, respectively. The AOA in Allende consists mainly of olivine, high-Ca pyroxene, anorthite, nepheline and low-Ca pyroxene on its periphery. It was observed that a cluster of fine equigranular sizes of olivine grains exists within the AOA (figs 3-4). The olivine cluster showed variation in Fe to Mg content. The Fe-rich olivine was generally found in along the periphery and interstitial sites of Mg-rich olivine grains. AOA's Ca, Al-rich phases include Ti-rich Al-Diopside (Ti-Al rich Ca-pyroxenes), anorthite, and nepheline. This assemblage of Al-rich minerals was termed as "Al-rich inclusions" by Hashimoto and Grossman (1987). The Al-inclusions were observed along and within olivine clusters. Many small rounds in shape cavities were observed within the grains as well. The cavities observed could be because of the sample preparation method, but Imai and Yurimoto (2002), after analysing the sample in a thin section, mentioned that they would have expected the shape of the cavity to be irregular and angular. Also, if the small round crystals did break during the sample preparation, then the remnant crystal could be observed in the cavities, which was not the case.

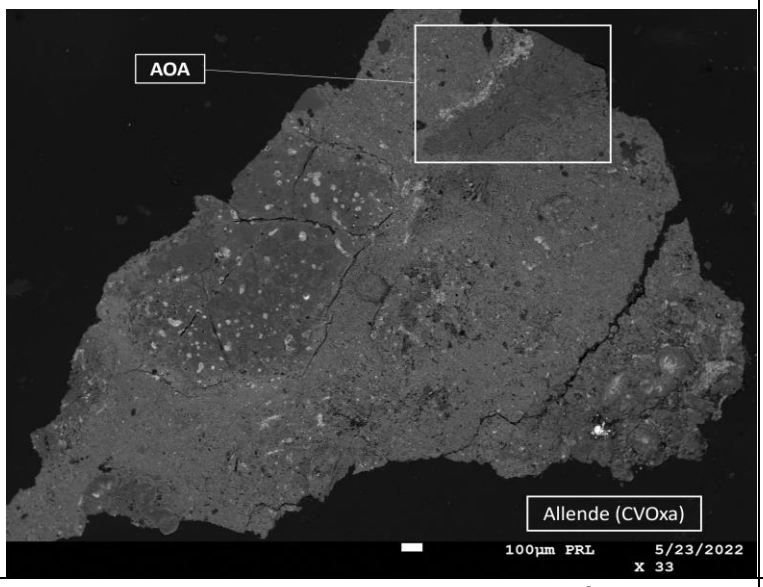


Figure 3(a). *Back-scattered Electron image of the Allende sample, taken at 30x. AOA in the top corner can be seen*

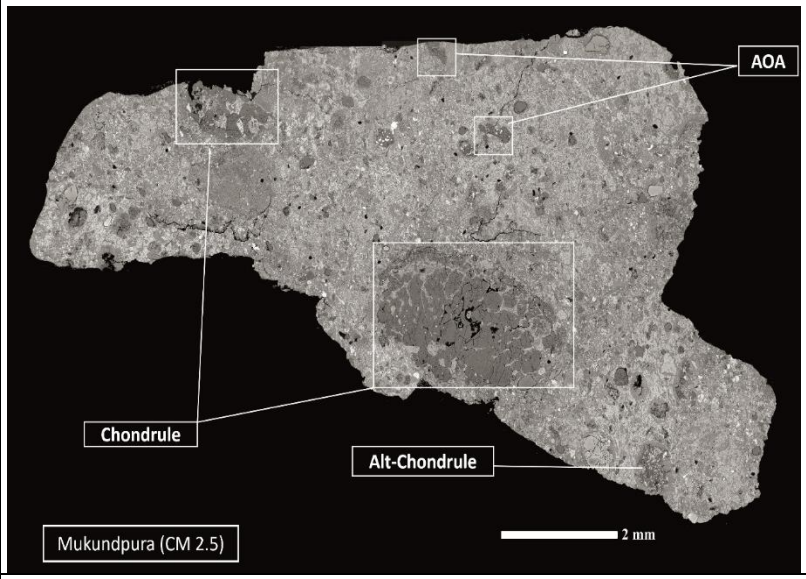
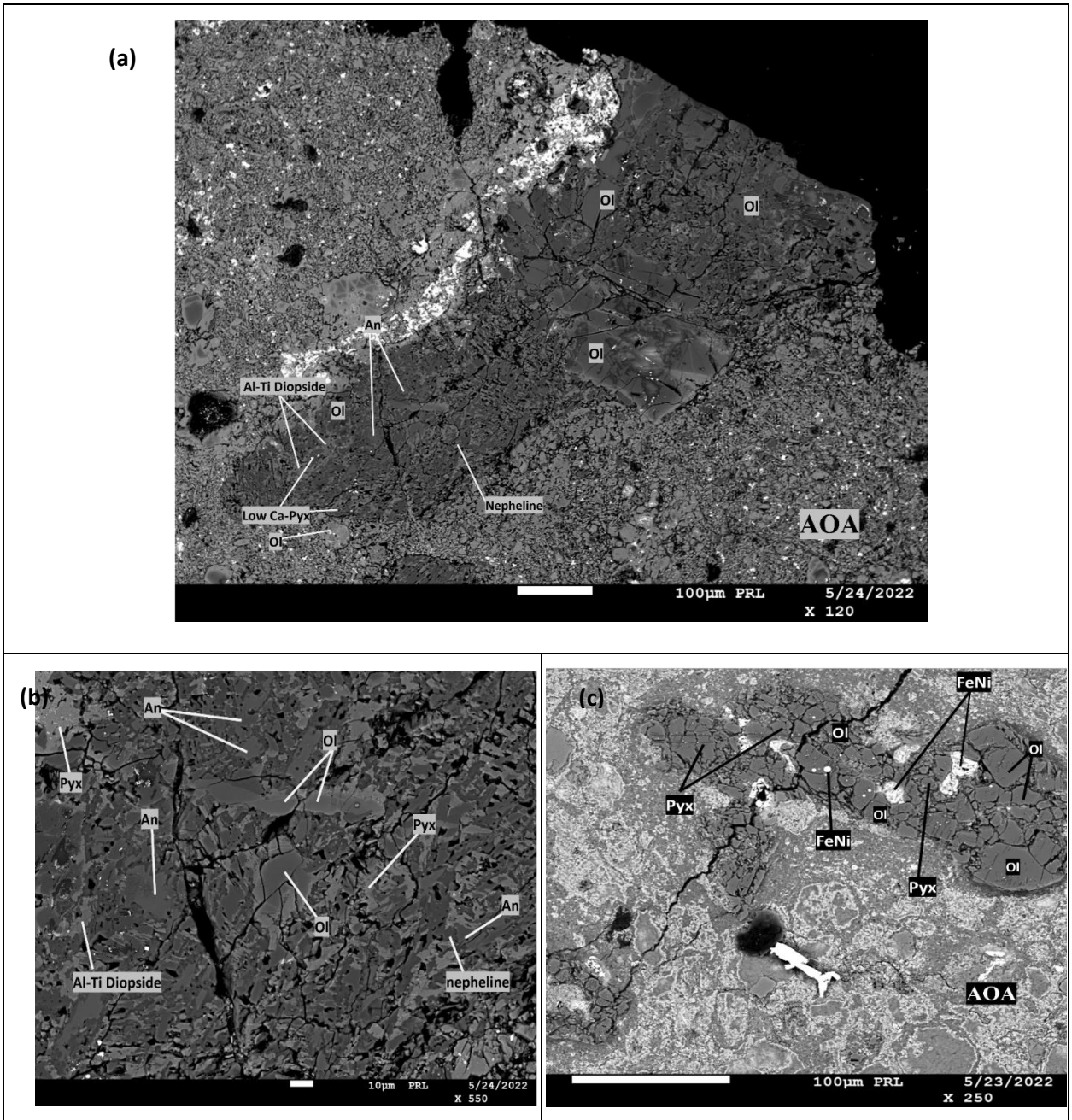


Figure 3(b). *Back-scattered Electron image of the Mukundpura sample. Processed after collaging individual images taken at 250x in Adobe Photoshop.*


(a)
Ol
Ol
An
Ol
Al-Ti Diopside
Ol
Low Ca-Pyx
Ol
Nepheline
AOA
100µm PRL
5/24/2022
X 120
(b)
An
Ol
Pyx
An
Pyx
Ol
An
Al-Ti Diopside
nepheline
10µm PRL
5/24/2022
X 550
(c)
FeNi
Ol
Ol
Pyx
Ol
FeNi
Ol
Pyx
AOA
100µm PRL
5/23/2022
X 250

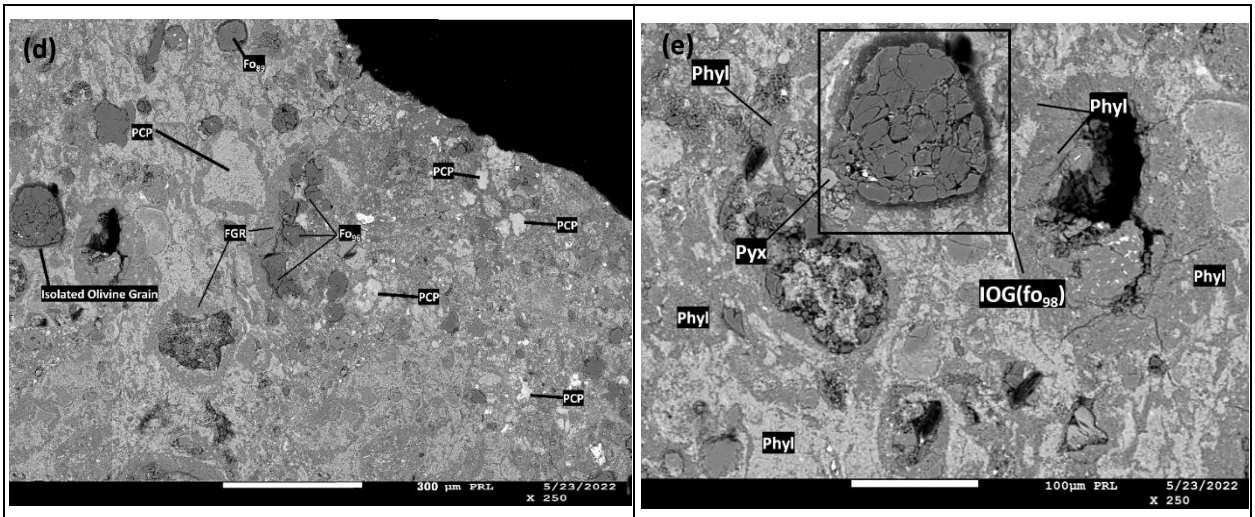


Fig 4: *(a) BSE image of AOA in Allende at 120x, (b) BSE enlarged image of AOA observed in Allende at 550x, olivine, pyx, Al-Ti-diopside, anorthite and nepheline can been seen ; (c) BSE image of AOA in Mukundpura at 250x, (d) and (e) phyllosilicates as fine-grained rims (FGR) around forsterite grains and PCP clumps in the matrix. (IOG-isolated olivine grains)*

The AOA observed in Mukundpura (CM 2.5) consists mainly of olivine grains and Mg-rich pyroxenes. It was observed as a very compact object. The AOAs are generally found dominantly as refractory inclusions in CM chondrites. MacPherson et al. (1988) have studied AOAs in detail only in Murchison, which explains the texture as Mg-rich olivine with refractory inclusions of CAIs and fine-grained phases spread within the cluster of forsterite grains, with their alteration products as phyllosilicates and carbonates. Although the AOA observed in Mukundpura was of compact type but showed no pores in their olivine grains as compared to Murchison, where leaching out of refractory phases from the aggregates of Olivine grains had left abundant pores. The AOAs and the matrix in the bulk Mukundpura chondrite showed abundant phyllosilicate minerals. The coarse components in Mukundpura other than AOA showed fine-grained matrix-like rims of phyllosilicates around them. Clumps of PCP (Poorly characterised phases) could also be seen in the matrix (fig 4 (d) and (e)).

Mineral Chemistry

Most of the olivine grains observed in AOAs in Mukundpura are nearly pure forsterite (**Fo_{96-97}**), whereas olivine in AOAs of Allende has composition from **Fo_{64-99}**, with peak touching around ~ **Fo_{80-85}**. However, some olivine grains are extremely rich in Fe (**Fa_{64}**). The representative compositions of olivine grains are shown in Table 1. The *low-Ca Pyroxene* (***$Fs_{1-4}Wo_{1-5}$***) contains (in wt%) 3-7 Al_2O_3 and 0.8-1.3 TiO_2, whereas *high-Ca, Ti- pyroxene* contains up to 9.2 wt% Al_2O_3 and <6 wt% TiO_2. The data is similar in the composition of pyroxene in AOAs in CV chondrite, as studied by Grossman and Steele (1976). Representative analyses are given in table 2. The pyroxene in Mukundpura AOA observed is mainly Mg-rich (***En_{96-91}*).** The plagioclase observed in

Allende-AOA is mainly anorthitic in composition (**An_{74}**)(table 3). A small grain of nepheline [(Na, K)AlSiO4] was also observed surrounded by anorthite inside the AOA (Allende). For Mukundpura, chondrules and other isolated olivine grains (IOG-**Fo_{98}** fig-4(e)) contain fine-grained rims of altered secondary phases- phyllosilicates.

Table 1. Representative electron microprobe analyses of olivine in AOAs

	#alln	#alln	#alln	#alln	#mkd	#mkd	#mkd	#mkd
SiO_2	39.97	38.5	36.42	35.47	40.03	40.8	41.69	37.9
TiO_2	0.1	0.07	0.28	0.214	0.029	0.006	0.02	0
Al_2O_3	0.04	0.105	6.82	1.84	0.135	0.126	0.03	0.72
Cr_2O_3	0.08	0.108	0.55	0.08	0.474	0.54	0.58	0.45
FeO	9.7	13.99	18.66	25.12	1.402	1.456	1.47	1.6
MnO	0.04	0.15	0.09	0.13	0.185	0.16	0.17	0.2
MgO	50.89	45.63	33.44	33.80	56.08	56.05	56.95	51.03
CaO	0.1	0.09	0.27	0.17	0.22	0.21	0.21	0.2
Na_2O	0	1.22	3.38	3.12	0	0	0	0
K_2O	0.04	0.008	0.04	0.018	0.002	0	0	0
Total	100.96	99.90	100.12	100.0	98.88	99.36	101.22	92.10
Fo %	**83.85**	**76.39**	**63.84**	**76.39**	**97.18**	**97.10**	**97.13**	**96.59**

Table 2. Representative electron microprobe analyses of low-Ca and high-Ca pyroxenes in AOAs

	#Alln px	#Alln px	#alln px	#alln px	#mkd px	#mkd px	#mkd px	#mkd px	#Alln cpx	#Alln cpx
SiO_2	57.83	53.07	54.68	45.67	58.51	49.22	55.47	57.24	43.9	45.14
TiO_2	0.46	0.55	1.06	0.64	0.185	0.16	0.13	0.12	4.79	5.38
Al_2O_3	2.77	4.01	5.8	2.91	0.91	2.42	0.98	0.39	9.23	8.9
Cr_2O_3	0.72	0.62	1.46	0.66	0.44	0.52	0.66	0.63	1.56	1.46
FeO	0.82	3.03	1.09	12.12	0.965	1.77	1.59	1.23	4.77	3.91
MnO	0.16	0.06	0.29	0.06	0.07	0.13	0.23	0.14	0.58	0.35
MgO	37.73	33.86	34.05	35.85	40.235	32.05	36.43	38.14	15.16	14.94
CaO	0.36	2.99	0.98	1.0	0.485	0.72	1.78	0.77	19.02	20.98
Na_2O	0.24	0.5	1.39	1.35	0	0	0	0	0.21	0.34
K_2O	0.02	0.14	0	0.01	0.005	0.005	0.01	0	0.04	0.1
Total	101.14	99.83	100.82	100.33	101.9	87.19	97.53	98.66	99.26	101.76
Fs %	**2.107**	**7.59**	**3.01**	**24.7**	**2.31**	**5.11**	**3.97**	**3.06**	**12.24**	**9.81**
Wo%	**0.92**	**7.49**	**2.71**	**2.04**	**1.16**	**2.08**	**4.49**	**1.91**	**48.83**	**52.67**

Table 3. Representative electron microprobe analyses of plagioclase and nepheline in AOA				
	Plagioclase		Nepheline (Na,K)$AlSiO_4$	
	wt%	*cations p.f.u.*	*wt %*	*Cations p.f.u.*
SiO_2	46.15	2.135	43.24	0.994
TiO_2	0.04	0.002	0.02	0
Al_2O_3	32.64	1.78	37.63	1.02
Cr_2O_3	0.07	0.03	0	0
FeO	1.048	0.041	0.61	0.012
MnO	0.036	0.001	0.05	0.001
MgO	0.934	0.064	0.31	0.011
CaO	14.632	0.725	2.18	0.54
Na_2O	4.932	0.442	16.84	0.751
K_2O	0.014	0.001	1.98	0.058
Total	**100.51**	**5.10**	**102.86**	**2.900**
	An~74.73%			

DISCUSSION

The AOAs are mostly irregular-shaped bodies in a chondritic meteorite made up of 3 main components: CAIs, ferromagnesium component, and FeNi metal inclusions. The CAIs in AOAs are composed of Al-Diopside, anorthite, spinel, and ± melilite, and ferromagnesium component composed of forsterite and low-Ca pyroxene.

According to Grossman and Steele (1976), the nebular gas condensed into perovskite, spinel, forsterite-olivine, and Al, Ti-pyroxenes between 1400 and 1600 K. Some of these grains then continued to equilibrate with the nebular vapour as it cooled to 900–500 K to form fayalitic olivine, nepheline, and sodalite. The subsequent accretion of these micron-sized grains created subspherical aggregates. Some of the clumps were covered in a coating of coarser-grained forsterite that condensed from the heated nebular gas prior to their aggregation into AOAs.

Hashimoto and Grossman (1987) later showed that in Allende AOAs, the Ca and Al- inclusions follow an alteration sequence and therefore concluded that the alteration took place in solar nebula after the aggregation of high-temperature condensates.

Our petrographic observations suggest the following formation sequence of minerals in AOAs:

spinel → melilite → Al-diopside ± spinel ± anorthite → forsterite + Fe, Ni-metal → low-Ca pyroxene.

This crystallisation sequence based on the thermodynamic calculations is given by Petaev and Wood, 1998. They showed that it is consistent with the thermodynamically predicted condensation sequence at $P_{total} = 10^5$ bar (fig-3,4).

Melilite, the primary mineral in CAIs, may or may not be present inside AOAs. If present, it is extensively replaced by fine-grained symplectite intergrowth of spinel, Al-diopside and ±anorthite. Low-Ca pyroxene is generally observed texturally on the periphery of AOAs.

The condensation temperatures of minerals are:

Minerals	*Condensation temperature (at 10^{-5}bar)*
Melilite	1503 K
Spinel	1371 K
Diopside	1323 K
Forsterite	1318 K
Anorthite	1293 K
Fe-Ni metal	1290 K
Orthopyroxene	1263 K

The condensation sequence is the same at a total pressure of 10^{-3} bar, but at temperatures of 1462 K and 1444 K, respectively, FeNi-metal condenses before forsterite (Ebel and Grossman, 2000). These findings lead us to the following conclusions: (i) AOAs are aggregates of gas-solid nebular condensates, perhaps including igneous CAIs; and (ii) AOAs were not thoroughly melted after aggregation and are potential evidence of high-temperature nebular reactions.

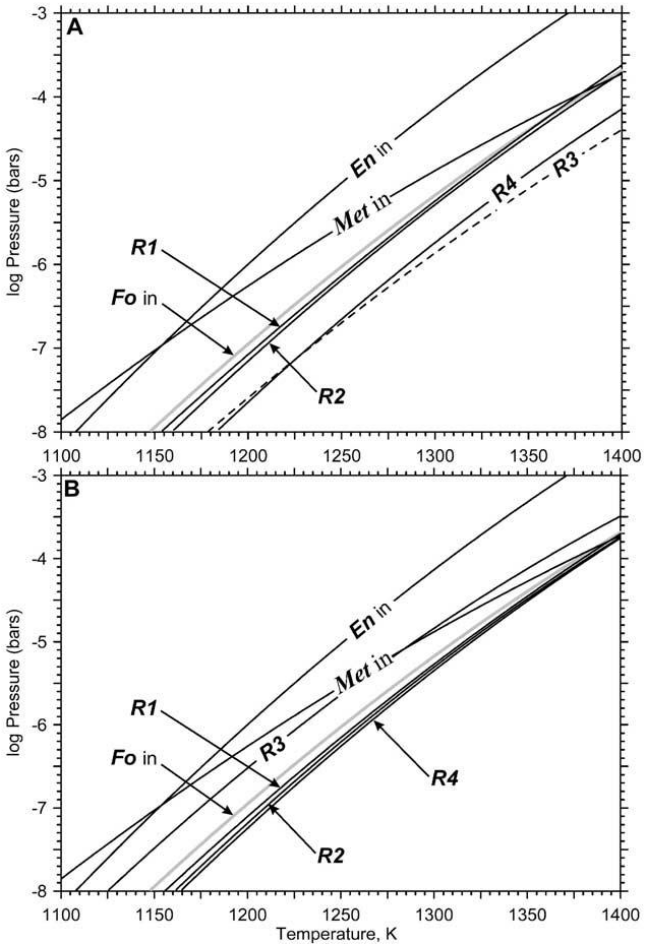


Figure 3: R1-R4- equilibrium condensation curves of minerals and phase boundaries of reactions based on the thermodynamic analysis of condensation reactions recorded in AOAs, after Krot et al (2004)

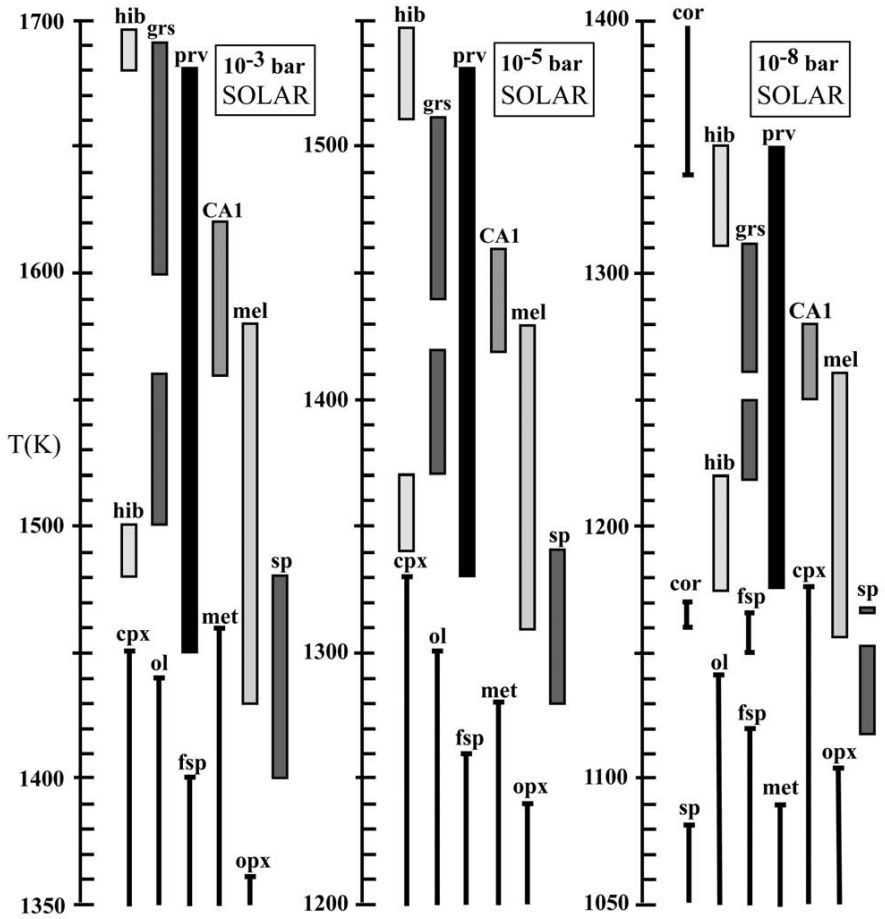


Figure 4 Appearance and disappearance temperatures of different minerals at different pressure (after Weisberg et al. 2004)

The condensation reactions recorded in AOAs are as follows:

The replacement of melilite by a fine-grained combination of spinel, Al-diopside, and anorthite that has been reported in AOAs can be understood by reactions of melilite with gaseous Mg and SiO:

$$\mathrm{Ca_2Al_2SiO_{7(g)(melilite)}} + 3SiO_{(g)} + 6\mathrm{H_2}O\ _{(g)} \rightarrow$$

$$\mathrm{CaMgSi_2O_{6(g)(diopside)}} + Mg\mathrm{Al_2O_{4(g)(spinel)}} \qquad \text{------(1)}$$

$$Ca_2Al_2SiO_{7(g)(melilite)} + 3SiO_{(g)} + Mg_{(g)} + 6H_2O_g \rightarrow$$
$$CaMgSi_2O_{6(g)(anorthite)} + CaMgSi_2O_{6(g)(diopside)} + 4H_{2(g)} \quad \text{------ (2)}$$

Spinel further gets corroded to various degrees and gets surrounded by anorthite and Al-diopside. The following reaction can explain the observation:

$$MgAl_2O_{4(g)(spinel)} + 2CaMgSi_2O_{6(s)(diopside)\ 2} + 0.5SiO_{(g)} +$$
$$0.5H_2O_{\ (g)} \rightarrow CaMgSi_2O_{6(g)(anorthite)} + MgSiO_{4g} +$$
$$0.5CaAl_2SiO_{6(s)(Al-diopside)} + 0.5HO_{2(g)} \quad \text{---------- (3)}$$

In some cases, instead of the appearance of anorthite with diopside, we observe a complete lack of anorthite and notice the Al-enrichment of diopside and Mg-rich pyroxene. The reaction involves spinel and melilite, and it proceeds until the disappearance of these minerals,

$$Ca_2Al_2SiO_{7(s)(melilite)} + MgAl_2O_{4(g)(spinel)} + 0.5SiO_{(g)} + Mg_{(g)} +$$
$$6H_2O_{(g)} \rightarrow \lfloor CaAl_2SiO_6 + 2CaMgSi_2O_6 \rfloor_{(\text{pyroxene})} + 4H_{2(g)} \quad \text{------- (4)}$$

However, the above reaction does not affect the modal mineralogy of CAIs and AOAs.

For forsterite- the primary component of an AOA is undoubtedly formed by direct condensation from nebular gas according to the reaction:

$$2Mg_{(g)} + 3SiO_{(g)} + 3H_2O_g \rightarrow Mg_2SiO_{4(s)(forsterite)} + 4H_{2(g)} \quad \text{-----(5)}$$

The presence of low-Ca pyroxene is generally observed texturally on the periphery of AOAs. Krot et al. (2004,2005) have proposed with the study of oxygen isotope that the low-Ca pyroxene replaces forsterite in outer portions of AOAs. And that the ^{16}O-poor low-Ca pyroxene shells around

^{16}O-rich CAIs must have accreted after remelting of forsterite grains at a relatively low nebular temperature in ^{16}O-poor gaseous reservoir during the chondrule formation. And the condensation sequence reaction at relatively low nebular temperatures is as follows:

$$\mathrm{Mg_2SiO_{4(s)}}(forsterite) + SiO_{(g)} + \mathrm{H_2}O_g \rightarrow \mathrm{Mg_2SiO_{6(s)}}(Low\text{-}Ca\ pyx) + H_{2(g)} \quad \text{---------- (6)}$$

Also, the presence of forsterite and Fe, Ni- metal must have directly condensed from nebular gas according to the reaction as follows:-

$$2Mg_{(g)} + SiO_{(g)} + 3\mathrm{H_2}O_g \rightarrow \mathrm{Mg_2SiO_{4(s)}} + H_{2(g)} \quad \text{---------- (7)}$$

$$\mathrm{Fe_{(g)} + Ni_{(g)} \rightarrow Fe, Ni_{(s)}} \quad \text{---------- (8)}$$

Other than these reactions supporting the observed minerals in AOAs, we could see multiple clear signs of aqueous alteration in Mukundpura chondrite. The majority of chondrules have phyllosilicates within them (which is a secondary mineral). However, some minerals survived, which were more resistant. Moreover, all the chondrules are surrounded by fine-grained phyllosilicate thick rims dominated by Fe-rich phases demonstrating abundant aqueous phases. Carbonates were also identified (~7%) embedded in the matrix, mostly calcite (CaCO3) (Ray and Shukla, 2018; Rudraswami et al., 2018), but Rudraswami et al., 2018 also confirms the presence of dolomite (CaMg(CO3)2), which is another sign of aqueous alteration. The presence of dolomite also may be possible evidence, as it was also found in other CM2 chondrites. There are apparently no shreds of evidence of thermal metamorphism.

Chapter-5

CONCLUSION

AOAs consist of forsterite grains surrounding Ca-Al-rich minerals. We observed a sequence of condensation reactions indicating the formation environment of these minerals. The minerals formed are of both high- and relatively low-temperature environments. AOAs are genetically related to CAIs and have chemical similarities to magnesium type I chondrule. The forsterite grains in AOAs in Allende are fayalitic in composition, whereas AOAs in Mukundpura are dominantly forsterite in composition ($fo_{>98}$). Al-Ti- diopside and anorthite indicate that the Ca-Al-rich minerals react in SiO and Mg-rich gas to produce such minerals (eq 1 to 5 and 7). The oxygen isotope study has shown that the CAIs and the primary minerals in AOAs have formed in ^{16}O-rich gaseous nebula. However, the presence of low-Ca pyroxene on the periphery of AOAs has condensed in a ^{16}O-poor gaseous nebula similar to the chondrules forming environment. It has been studied that the bulk compositions of the AOAs are moved towards chondrules by the formation of low-Ca pyroxene, which results from the reaction of the AOAs with gaseous SiO and from the melting of silica-rich dust that has accumulated around the AOAs (eq 6). This process also suggests potential mechanisms for converting refractory materials into chondrules or chondrule precursors. The rare occurrences of low-Ca pyroxene in AOAs may suggest that either the AOAs were separated from the hot nebular gas before low-Ca pyroxene condensation took place or that the reaction between forsterite and gaseous SiO was kinetically hindered.

Future Scope:

The isotope study, especially O-isotope, is critical to understanding the environmental condition these AOA have formed. Other isotopes like Al-Mg can be used to determine the formation age.

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